

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

18MEE11 - FINITE ELEMENT METHODS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Briefly explain the basic steps involved in the FEM for stress analysis of elastic solid body. | 10 | L*2 | 1 | 1 |
| b) Write a short note on
i) Principle of minimum potential energy
ii) Rayleigh – Ritz Method | 10 | L2 | 2 | 1 |
| 2. a) Explain the concept of plane stress and plane strain with neat sketch. | 4 | L2 | 1 | 1 |
| b) Discuss the types of elements (1D, 2D and 3D) based on geometry. | 6 | L2 | 1 | 1 |
| c) Using principle of minimum potential energy, determine the nodal displacement of spring system as shown in Figure Q2 (c). | | | | |

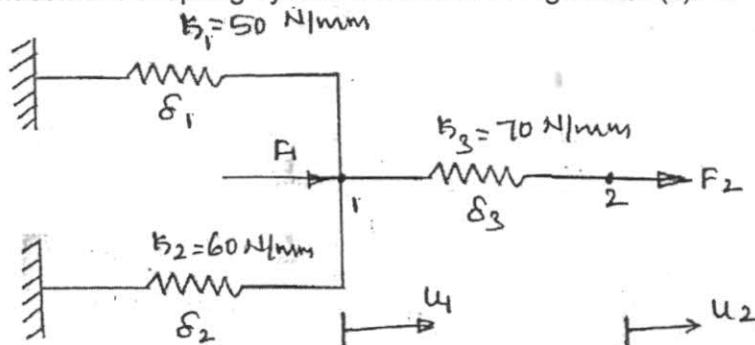


Figure Q2 (c)

- | | | | | |
|---|----|----|---|---|
| 3. a) Derive the stiffness matrix for 1 Dimensional bar element using direct stiffness method. | 10 | L2 | 2 | 1 |
| b) A bar of length L, cross sectional area A and modulus of elasticity E, is subjected to distributed load $q=cx$, where c is a constant as shown in Figure Q3 (b). Determine the displacement at the end of the bar using Rayleigh – Ritz method. | 10 | L3 | 1 | 1 |

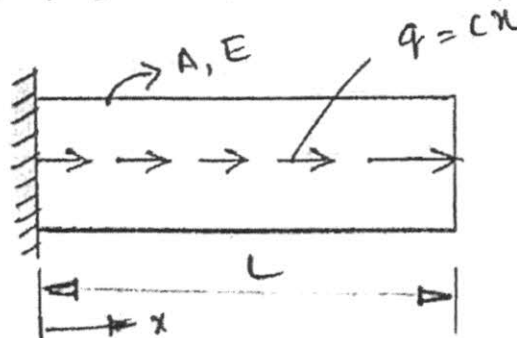


Figure Q3 (b)

10	L3	2	1
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Unit – II

4. a) Derive the shape function for one dimensional bar element in Cartesian coordinate/Global coordinate system. 10 L3 3 1
 b) Derive an Elemental stiffness matrix of 2D truss element. 10 L3 4 1
5. a) Write a note on
 i) Selection of nodes in finite element method
 ii) Convergence requirements 10 L2 3 1
 b) Consider the bar shown in Figure Q5 (b). An axial load $P=200 \times 10^3 \text{ N}$ is applied as shown in figure using penalty approach. Determine the following:
 i) Nodal Displacement ii) Stress in each material

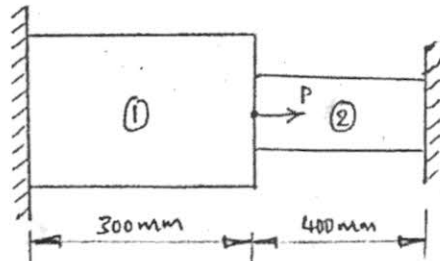


Figure Q5 (b)

$$A_1 = 2400 \text{ mm}^2$$

$$E_1 = 70 \text{ GPa}$$

$$A_2 = 600 \text{ mm}^2$$

$$E_2 = 200 \text{ GPa}$$

10 L3 4 1

6. a) Explain the different types of interpolation models and write the polynomial forms of 1D, 2D and 3D Elements. 10 L3 3 1
 b) Derive the shape functions in local coordinate system for 1 dimensional bar element. 10 L3 4 1

Unit – III

7. a) Derive the shape function for two noded beam element. 10 L3 5 1
 b) Determine the maximum deflection in the uniform cross section of cantilever beam shown in Figure Q7 (b) by assuming beam as single element. Take $E=7 \times 10^9 \text{ N/m}^2$, $I = 4 \times 10^{-4} \text{ m}^4$.

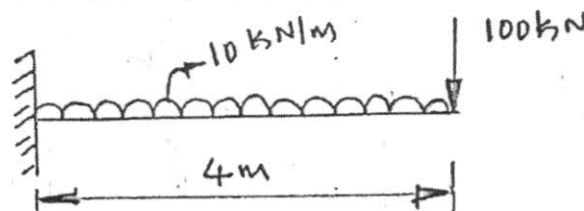


Figure Q7 (b)

10 L3 5 1

8. Determine the vertical deflection and slopes, at point 2 and 3, using beam elements for the structure as shown in Figure Q8. Also determine the deflection at the center of the portion of the beam carrying UDL.

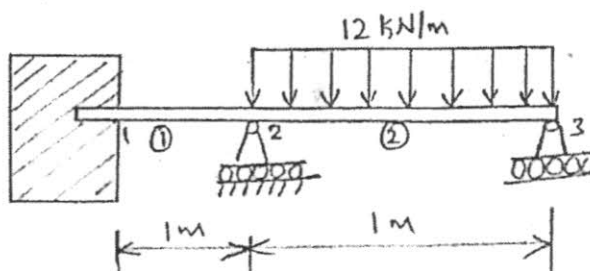


Figure Q8

$$E = 200 \text{ GPa}$$

$$I = 4 \times 10^{-6} \text{ m}^4$$

20 L3 5 1